

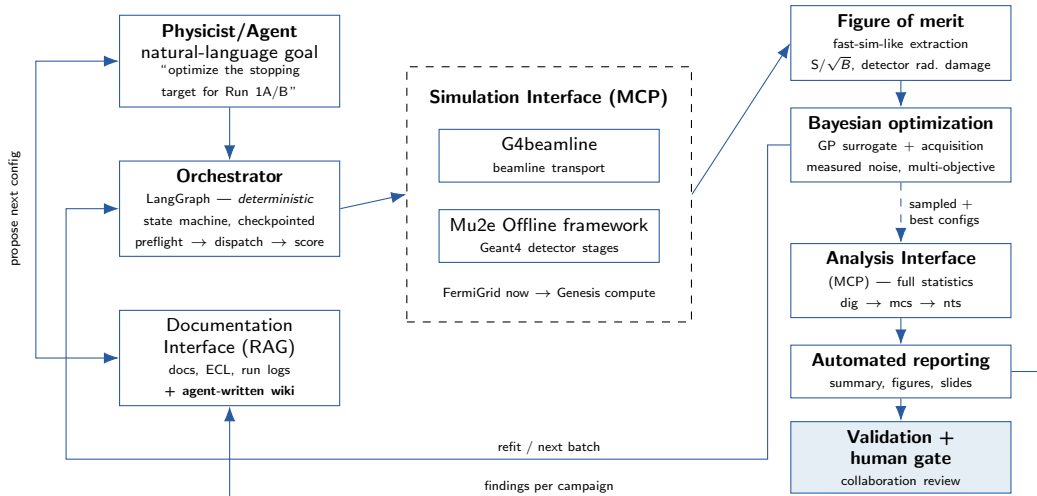
Mu2e Genesis: Agentic Experiment Optimization

Workflow (i) — activities and near-term plan

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Agentic Experiment Optimization — the loop

Prompt-to-configuration: simulate, analyze, score, decide — with a human gate.



Activities and owners (1/2)

Surrogate model optimization

(Yuri)

- ▶ Bayesian optimization: GP surrogate + multi-objective acquisition (sensitivity vs. detector radiation damage)
- ▶ Needed: multi-fidelity, transfer between scenarios
- ▶ Explore BO models for higher dimensionality: TuRBO, SAASBO, input warping

LangGraph orchestration

(Yuri / Ray / Rob)

- ▶ One graph run = one evaluation; checkpointed, resumable
- ▶ **Today: LangGraph + Bayesian optimization, prompted and orchestrated by Claude** — staying within Claude is a viable option
- ▶ Needed: expose through MCP, log traces
- ▶ Parameterize scenarios: beam intensity, solenoid field scale, arbitrary geometry configuration

Fast sim scoring, Run1A/B

(Michael)

- ▶ Today's only scoring path: $S/\sqrt{B}$ + radiation damage, no full reconstruction
- ▶ Needed: improve fast-sim performance; revisit Run 1B; agreement with full sim
- ▶ Explore other fast-sim options: beam optics, detector response

Activities and owners (2/2)

Full-sim validation (dig → mcs → nts)

(Sophie / James)

- ▶ Full chain, full statistics: best configs + every ~10th point, run async
- ▶ Needed: calibration vs. the fast score; evidence bar and sign-off

Infrastructure — MCP, LLM, computing

(Ray / Rob)

- ▶ MCP servers, agent skills, model endpoints, compute
- ▶ FermiGrid now → Genesis compute; token + CPU accounting

Documentation interface

(Cole / Simon)

- ▶ Search over docs, ECL, run and configuration logs
- ▶ Needed: context for the optimizer — which constraints are real

G4beamline

(Yuri)

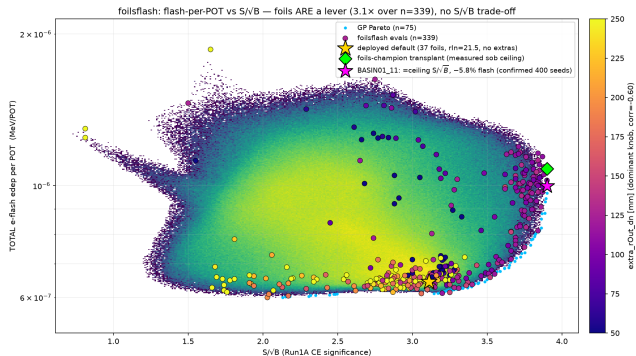
- ▶ MCP server prototype demonstrated — **a proposal commitment** (App. 5)
- ▶ Needed: integrate into the existing workflow

Automated reporting

(Rob / Yuri)

- ▶ Campaign drains → summary, figures, slides and write-up generated automatically
- ▶ Needed: regenerate on new rows; every number traceable to a leaderboard row
- ▶ **Write-back**: every campaign appends what it learned to the documentation interface — an agent-maintained wiki (Karpathy pattern); today hand-written per session

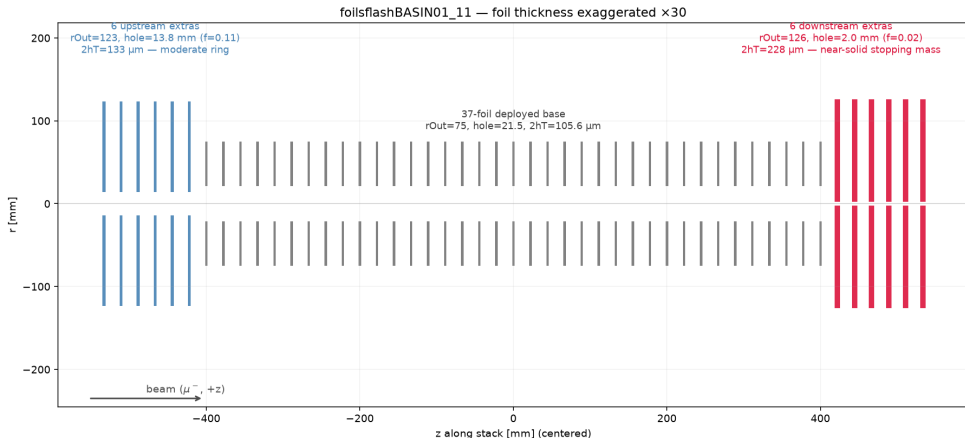
Status — the loop runs end-to-end



design	$S/\sqrt{B}$	rad. damage
deployed default	3.11	6.45×10^{-7}
ff11R00_07 (champion)	3.31	6.26×10^{-7}
ff14R01.00 (exploit)	3.84	8.14×10^{-7}
BASIN01.11 (confirmed)	3.90	1.00×10^{-6}

- ▶ **339** evaluations map a clean Pareto front
- ▶ Champion beats deployed on **both** axes: +6% $S/\sqrt{B}$ at -3% damage
- ▶ Best $S/\sqrt{B}$ **3.90 vs 3.11 deployed** = +25%, confirmed reproducible (400-seed)

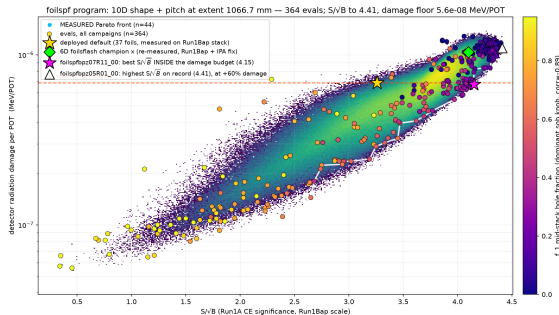
What the optimizer built — the $\max\text{-}S/\sqrt{B}$ target



Two routes to high $S/\sqrt{B}$: a thin **off-beam ring**, or thin **in-beam** material that degrades the beam to stop more muons. Here: 6 upstream extras (moderate ring) + 6 downstream (near-solid stopping mass) around the 37-foil deployed base.

Since last time — the whole target, not just the extras

Second campaign line: throw away the hand-designed 37-foil base and let BO design **all 49 foils** — outer radius, thickness and central hole as smooth profiles, plus the foil spacing. Everything stays inside the **deployed envelope**. At the *deployed damage budget* this gives +27% $S/\sqrt{B}$ (4.15 vs 3.26); unconstrained, the record is **4.41** = +35%.

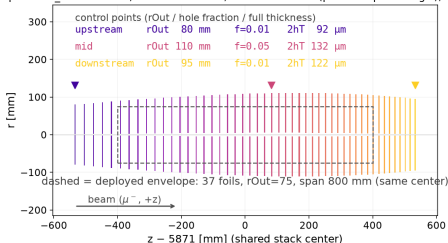


design	$S/\sqrt{B}$	rad. damage
deployed default	3.26	6.85×10^{-7}
bpz07R11_00 (deliverable)	4.15	6.70×10^{-7}
bpz05R01_00 (record)	4.41	1.10×10^{-6}

- ▶ **366** evaluations, twelve campaigns, **zero failures**
- ▶ The deliverable beats deployed on **both** axes: +27% $S/\sqrt{B}$ at -2% damage. **63** designs do this
- ▶ **43** measured Pareto points; damage frontier spans **23×**
- ▶ The old 4.00 plateau was an **acquisition artifact**: a picker aimed at the damage budget beat it in 4 evaluations

The max- $S/\sqrt{B}$ design — and how we knew to stop

foilsfpbz05R01_00 — 49 foils, extent 1067 mm, zmid +83.2 mm (pitch ramped along z), thickness $\times 30$



4.41 — the unconstrained record. A nearly **solid** stack (bore ≤ 5 mm) as a gentle barrel, rOut 80 $\rightarrow$ 110 $\rightarrow$ 95 mm, foils **packed downstream**. Nothing a human would have drawn.

Knowing when to stop is itself a result. Acquisition “saturation” means the *frontier* stopped moving — not that the space is mapped. Three dedicated max- $S/\sqrt{B}$ rounds:

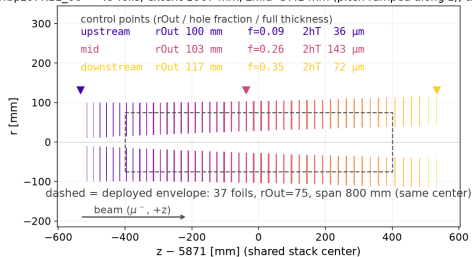
4.33 $\rightarrow$ 4.37 $\rightarrow$ **4.41** $\rightarrow$ 4.41

two $\sim 5\sigma$ steps, then flat

The third round: 40 evaluations, **nothing above 4.41** — and the optimizer spent a pick **re-measuring the record**, reproducing it exactly. Ridge mapped, line closed.

The design we would actually deploy — +27% for free

foilsfpbpz07R11_00 — 49 foils, extent 1067 mm, zmid -37.1 mm (pitch ramped along z), thickness x30



The record on the previous slide buys +35% by spending +60% more detector damage. **This one is free.**

	$S/\sqrt{B}$	rad. damage
deployed default	3.26	6.85×10^{-7}
bpz07R11_00	4.15	6.70×10^{-7}
difference	+27%	-2%

Better on both axes than the target now in the beam — **63** of 366 designs are, and this is the best of them.

The shape: a near-cylindrical stack, rOut 100 $\rightarrow$ 103 $\rightarrow$ 117 mm, with a **gradually opening bore** ($f = 0.09 \rightarrow 0.26 \rightarrow 0.35$) and the mass concentrated mid-stack (36/143/72 μm). Mass off-axis where it stops muons, beam core through the hole instead of into the detector.

Phase I milestones — where Workflow (i) stands

9 months from **1 Aug 2026**; go/no-go at month 6 (Jan 2027). **We are in month 1.**

window	dates	milestone (Workflow (i) parts)	status
Months 1–2	Aug–Sep 26	Deploy interfaces: Simulation, Analysis, Documentation End-to-end agentic chain: sim → analysis → FoM	pending done
Months 3–4	Oct–Nov 26	Exercise optimization on campaigns; collect metrics	ahead
Months 5–6	Dec–Jan 27	Compile go/no-go results; all traces logged	traces = gap
Months 7–9	Feb–Apr 27	Harden interfaces; package artifacts for Genesis	—

- ▶ One month in: the **science milestone of months 3–4 is already met**; interfaces are the open item
- ▶ Proposal §4 anticipates this: *“initial workflow prototyping can use simplified interfaces while production versions are improved”*

Decision gate metrics — Workflow (i), proposal §6

(a) *“Closed-loop sim–analysis–optimization (agent-orchestrated search with Bayesian optimization for local sampling) demonstrated on ≥ 2 distinct external-constraint scenarios (e.g. half beam intensity, 10% reduced fields), each initiated by a natural-language prompt and completed within one week (vs. months for the manual baseline), with full traces logged”*

- **Started.** Prompt → new mode + 20 evaluations, saturation confirmed by a second campaign — a scenario answers **within a day** (bar: one week)
- Remaining: the two **external-constraint** scenarios — reduced / absent DS field, half beam intensity — and **formal trace logging**

(b) *“for a Run-1a scenario (no full cosmic ray veto coverage), starting from the collaboration’s existing manually-optimized baseline, agent-produced configuration achieves improved $R_{\mu e}$ sensitivity, with improvements documented for collaboration review”*

- **Demonstrated twice, independently** — both lines beat the manually-optimized baseline on **both** axes:
 - ▶ extras-only line: +6% $S/\sqrt{B}$ at −3% damage; +25% at the confirmed ceiling
 - ▶ **full-target profile line (new)**: +27% $S/\sqrt{B}$ at −2% damage, **63** such designs; +35% at the mapped ceiling
- To finish: **prove it with full sims** (dig → mcs → nts) on the best configs; scope by background regime; document for review

Backup

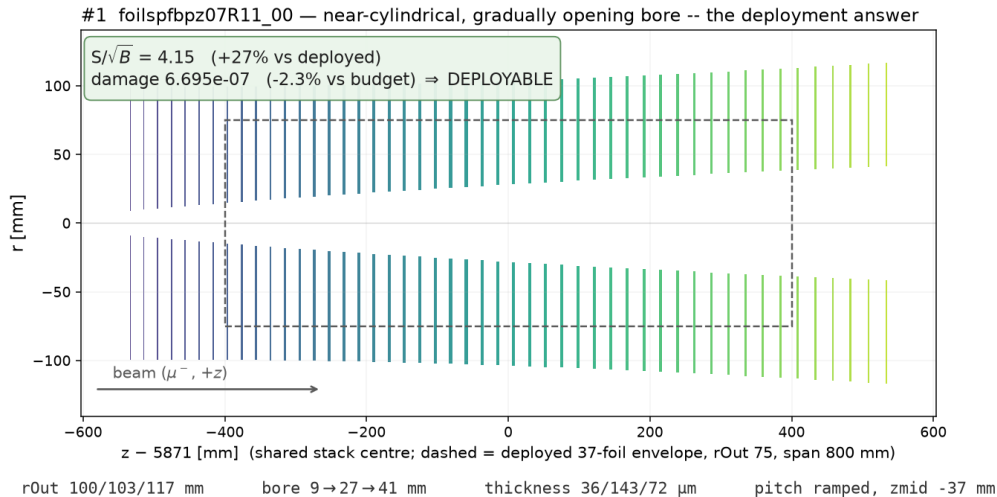
The five best deployable designs

Every design here is measured **at or below the deployed damage budget** — unlike the unconstrained record and the five that follow, these could go in the beam. All five come from bpz07, the one campaign whose acquisition was aimed *at* the constraint.

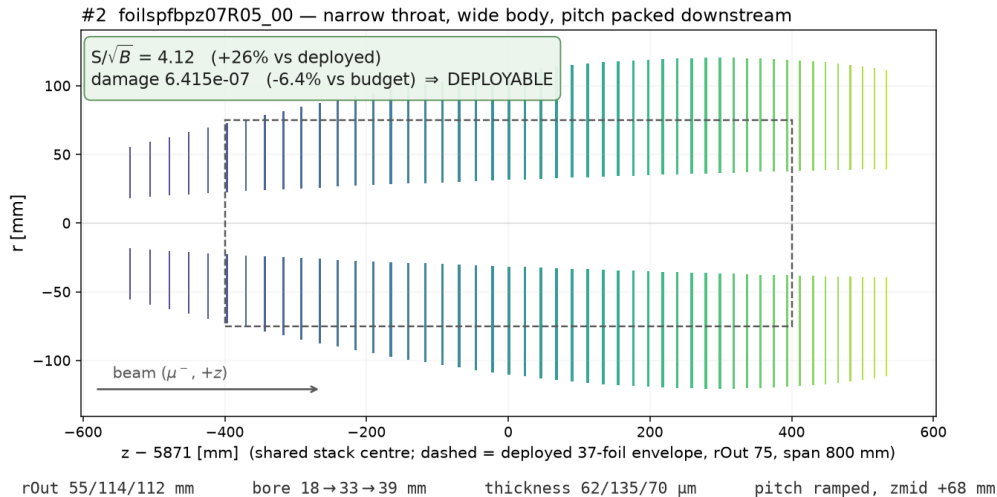
#	design	$S/\sqrt{B}$	vs deployed	damage	shape
1	bpz07R11_00	4.15	+27%	−2%	near-cylindrical, gradually opening bore
2	bpz07R05_00	4.12	+26%	−6%	narrow throat, wide body, packed downstream
3	bpz07R00_01	4.11	+26%	−6%	narrowest throat, uniform pitch
4	bpz07R09_00	4.11	+26%	−2%	widest downstream bore ($f = 0.52$)
5	bpz07R08_00	4.08	+25%	−9%	solid upstream face; cheapest damage

- ▶ The previous best deployable design was **4.00**, and three campaigns had failed to beat it — **all five of these do**
- ▶ A **spread of shapes**, not one recipe: upstream rOut 41–100 mm, bore fractions 0.02–0.33, pitch uniform to strongly ramped
- ▶ Every one of the five has **damage to spare** (−2 to −9%), so none is scraping the constraint to get its score

Deployable #1 — bpz07R11_00, $S/\sqrt{B} = 4.15$

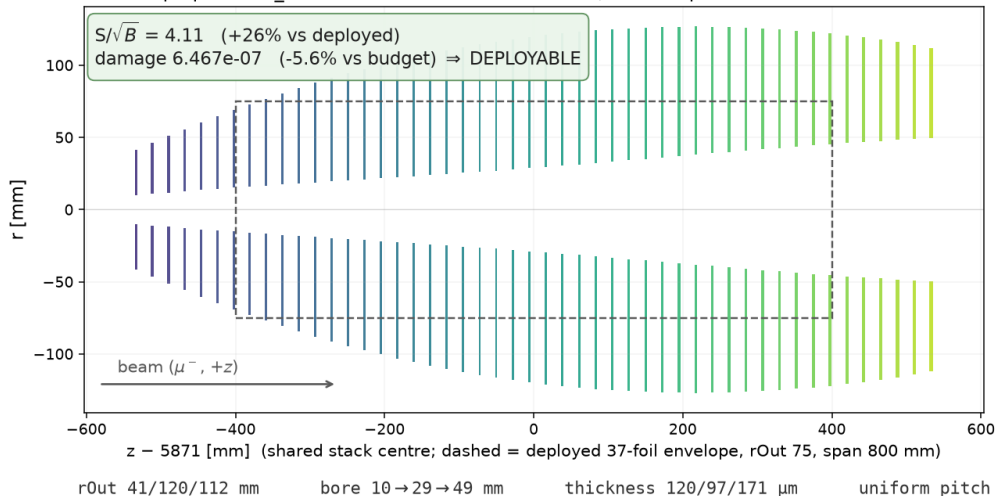


Deployable #2 — bpz07R05_00, $S/\sqrt{B} = 4.12$

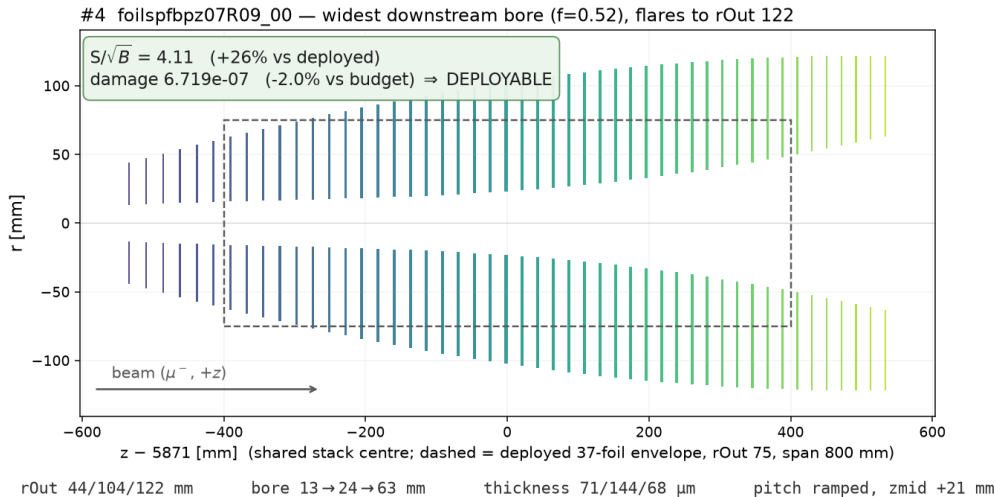


Deployable #3 — bpz07R00_01, $S/\sqrt{B} = 4.11$

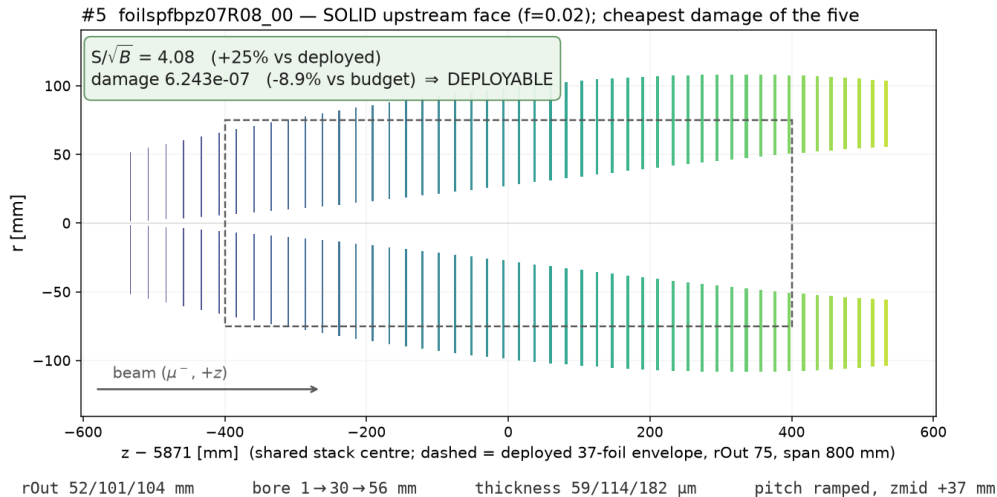
#3 foilspfbpz07R00_01 — narrowest throat of the five, uniform pitch



Deployable #4 — bpz07R09_00, $S/\sqrt{B} = 4.11$



Deployable #5 — bpz07R08_00, $S/\sqrt{B} = 4.08$



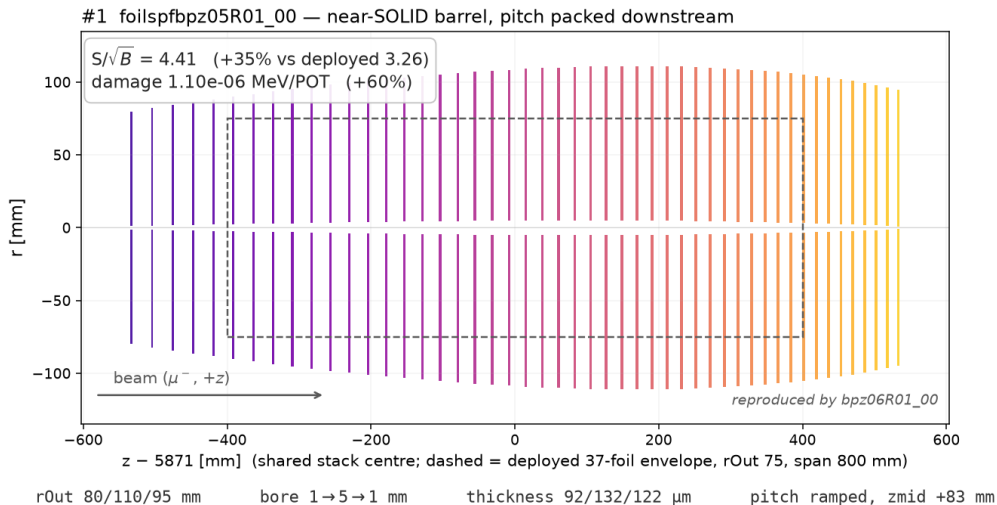
The five best designs OVER budget — one ridge top, five shapes

The optimizer's five best distinct targets score **4.38–4.41** — a spread of 0.03, i.e. $\sim 5\sigma$ total — yet look nothing alike. That degeneracy is the evidence the ceiling is mapped: many roads reach the same height, and none goes higher.

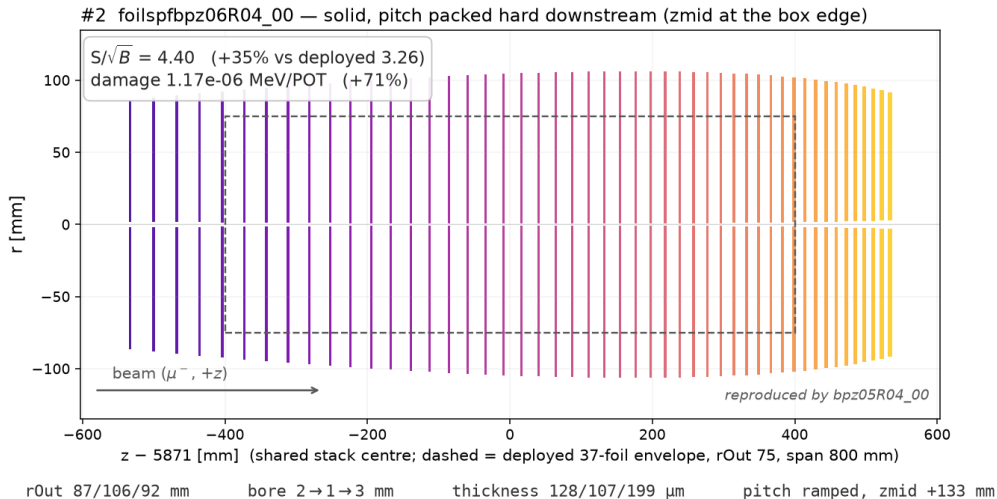
#	design	$S/\sqrt{B}$	damage	shape
1	bpz05R01_00	4.41	+60%	near-solid barrel, pitch packed downstream
2	bpz06R04_00	4.40	+71%	solid, pitch at the box edge
3	bpz05R04_03	4.38	+60%	small bore, very thick downstream foils
4	bpz05R08_00	4.38	+56%	modest bore, <i>uniform</i> pitch
5	bpz06R11_00	4.38	+53%	wide-mouth funnel bore, uniform pitch

- ▶ **Three of the five were measured twice** (different campaigns, independent seeds) and reproduced — the ridge top is real, not a set of lucky draws
- ▶ Two opposite hole strategies — **shut the bore** (#1–3) vs. **open a funnel** (#5) — reach the same score
- ▶ All five are **damage-unconstrained**; the deployment answer is bpz07R11_00 at **4.15** *inside* the deployed budget

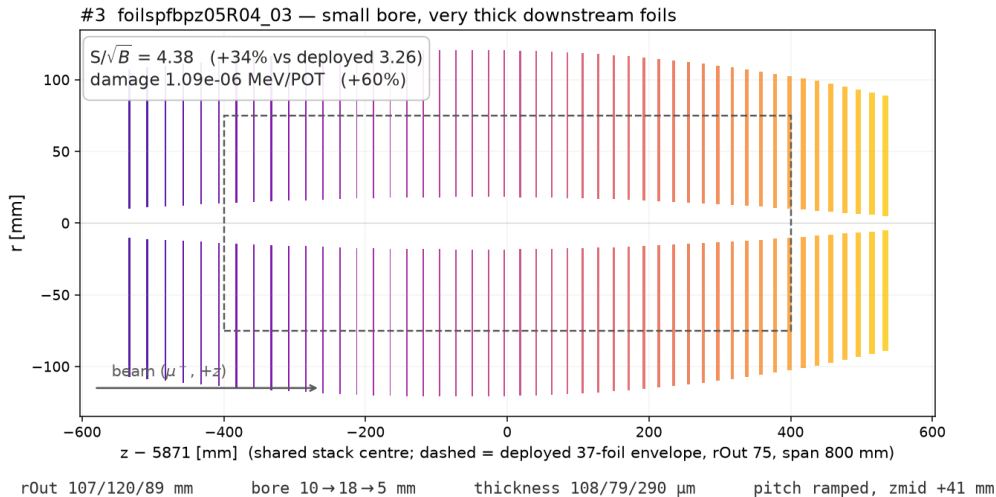
#1 — bpz05R01_00, $S/\sqrt{B} = 4.41$ (the record)



#2 — bpz06R04_00, $S/\sqrt{B} = 4.40$

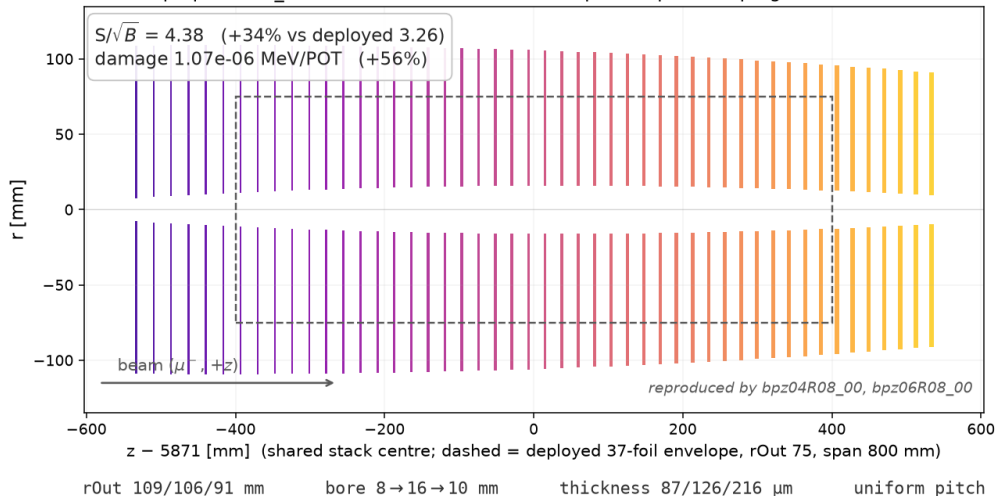


#3 — bpz05R04_03, $S/\sqrt{B} = 4.38$



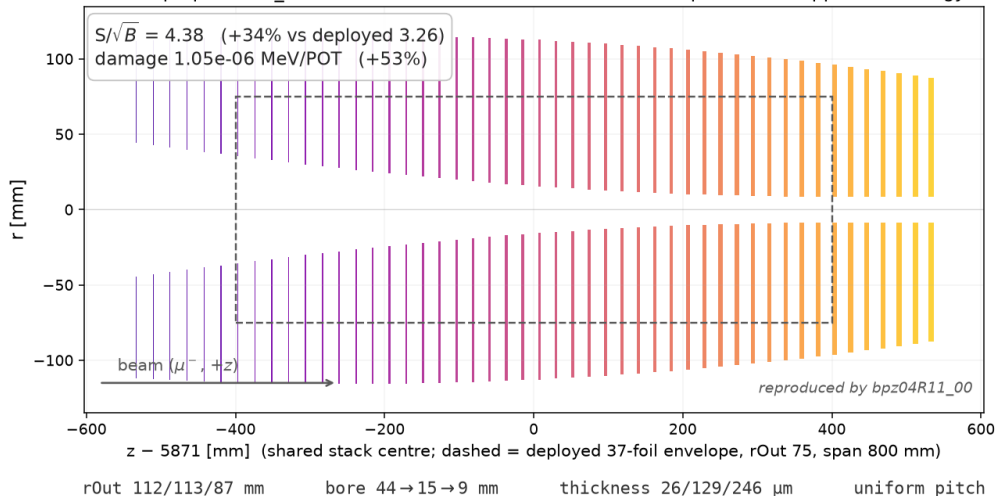
#4 — bpz05R08_00, $S/\sqrt{B} = 4.38$

#4 foilspfbpz05R08_00 — modest bore at UNIFORM pitch -- pure shape gain



#5 — bpz06R11_00, $S/\sqrt{B} = 4.38$

#5 foilspfbpz06R11_00 — wide-mouth FUNNEL bore at uniform pitch -- the opposite strategy



What “GP surrogate + multi-objective acquisition” means

GP surrogate — the map. Each real evaluation costs ~ 150 CPU-hours, so we cannot try many designs. The Gaussian Process looks at every design tried so far and draws a map of the whole space: for a design we have *not* tried it predicts the answer **and how unsure it is**. Near measured points the prediction is sharp; far away it is foggy. *The fog is the useful part.*

Acquisition — the rule for where to go next. Something has to choose which 10 designs to actually simulate next. The acquisition function trades off **exploit** (go where the map predicts good results) against **explore** (go where the fog is thick — we might be missing something).

Multi-objective — why there is no single “best”. We want two things at once: more signal, less detector damage. They fight each other, so there is no single winner — there is a **set** of best compromises, the **Pareto front** (the map on the status slide). Our acquisition function (qNEHVI) picks the designs expected to push that frontier *outward* the most, rather than chasing one number.